Module 4
Software Project Management

Syllabus

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Project:

Some dictionary definitions:

- “A specific plan or design”
- “A planned undertaking”
- “A large undertaking e.g. a public works scheme”
- A project is a temporary endeavor undertaken to create a unique product, service or result.
- A project is defined as a temporary organization that is created for the purpose of delivering one or more business products according to an agreed business case.

Characteristics of Projects

- Non-routine tasks are involved
- Planning is required
- Specific objectives are to be met or specified product is to be created
- The project has a predetermined time span
- Work is carried out for someone other than yourself
- Work involves several specialisms
- People are formed into a temporary work group to carry out the task
- Work is carried out in several phases
- The resources that are available for use on the project are constrained
- The project is large or complex

Project Parameters

- **Scope**: What is included?

  What is not included?

- **Quality:** Product, Process, People

- **Cost:** Project cost, Customer budget

- **Time**

- **Resources:** People, Equipment , Physical facilities

Software Projects V/S Other Engineering Projects

- **Invisibility**: When a physical artifact such as a bridge or road is being constructed the progress being made can actually be seen.

With software, progress is not immediately visible.

- **Complexity:** Software products contain more complexity than other engineered artefacts.
- **Conformity:** Software developers have to confirm to the requirements of human clients.
- **Flexibility:** Software project has the characteristics of changing its code at any time and can produce the expected result.

Software systems are particularly subject to change.

Contract management:

- Normally in the project the users and developers of new software work for the same organization.
- However, increasingly organizations contract out ICT development to outside developers.
- Here the client organization will often appoint a ‘Project Manager’ to supervise the contract who will delegate many technically oriented decisions to the contractor.
- Thus, the project manager will not worry about estimating the effort needed to write individual software components as long as the overall project is within budget and on time.

Activities Covered by Software Project Management:

1.Feasibility Study:

- Is project technically feasible and worthwhile from a business point of view?
- Investigation to decide whether a prospective project is worth starting

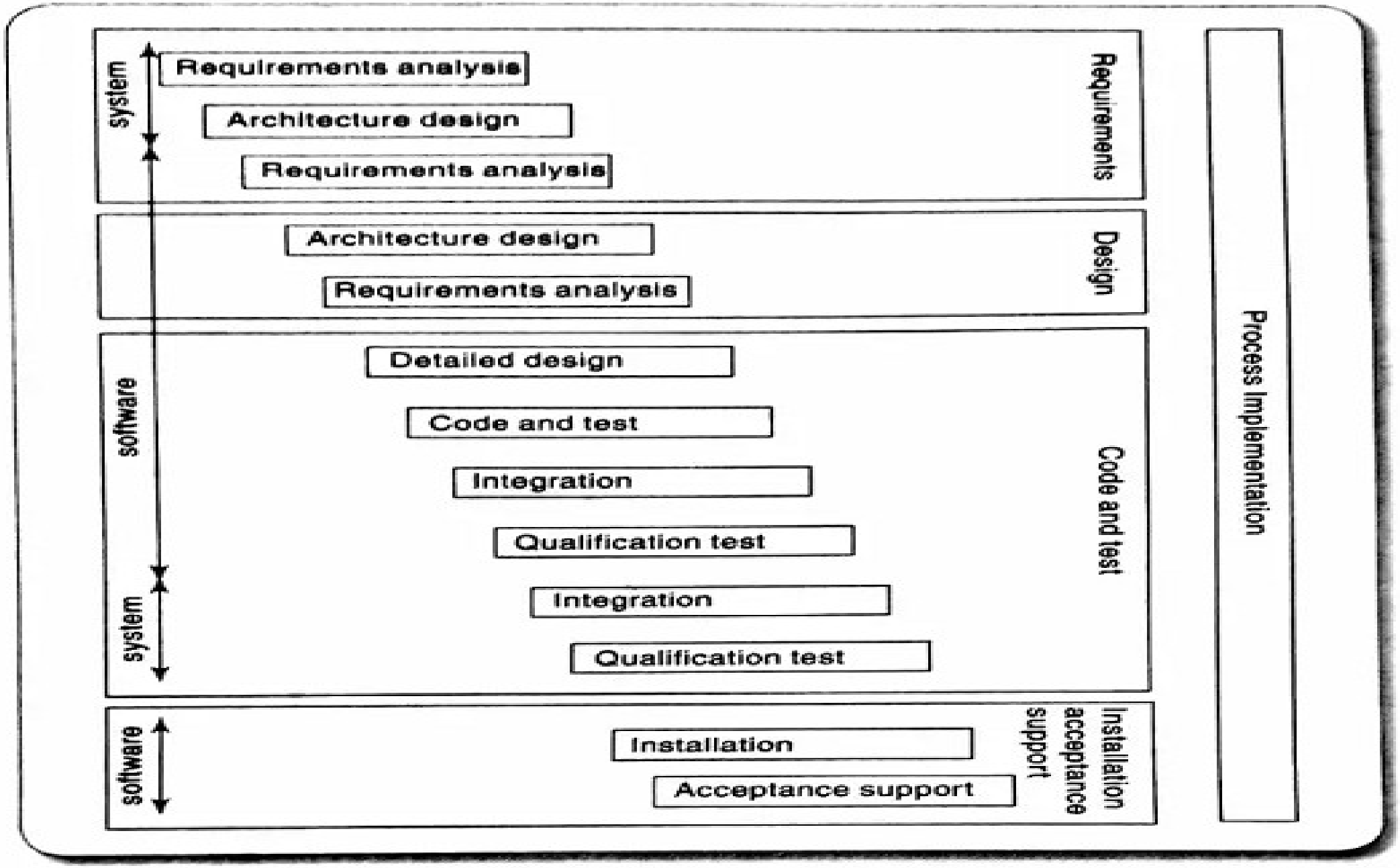
2.Planning:

- Only done if project is feasible
- In short it is defined as ‘deciding what is to be done’?
- However, for a large project, we should not do all our detailed planning right at the beginning
- More detailed planning of the later stages would be done as they approached.

3 Project Execution:

- This is the final stage of project, which meant to put the build system to work or operate under the suitable environment.
- Implement the plan but it may be changed as we go along with the project.
- The execution of a project often contains design and implementation of sub phases.

Software Development Life Cycle/ Stages of Project Life Cycle



Requirement Analysis:

- Requirements gathering which establishes what the potential users and their managers require of the new system.
- It could be relating to a function that the system should do something.

Architecture Design:

- Diagrammatic representation of the project which we going to do
- The components of the new system that fulfill each requirement have to be identified.
- Existing components may be able to satisfy some requirements and new components will have to be made.
- These components are not only software; they could be new hardware or work processes. Although software developers are primarily concerned with software components

Detailed Design:

- Each software component is made up of a number of software units which can be separately coded and tested.
- The detailed design of these units are carried out separately.

Code and Test:

- refers to writing the code for each software unit. Initial testing to debug individual software units would be carried out at this stage.

Integration:

- The components are tested together to see if they meet the overall requirements.
- It involves combining the different software components or testing the software elements of the system in conjunction with the hardware platforms and user interaction.

Qualification Testing:

- The system, including the software components has to be tested carefully to ensure that all the requirements have been fulfilled.

Installation:

- This is the process of making the new system operational.
- It includes activates such as setting up data, setting up system parameters, installing on operational hardware platforms, user training etc.

Acceptance Support:

- Resolves the problems of newly installed system, includes correction of any errors, implement the agreed extensions and improvements.

Plans, Methods and Methodologies:

Plan:

- The controlling document for the software project.
- Specifies the technical and managerial approaches to develop software project.

Method:

- It is a procedure or process for attaining an object.
- Such as a systematic procedure, technique or mode of inquiry employed by or proper to a particular discipline or a systematic plan followed in presenting material for instruction.
- In other words, a method refers to a single action, tool, technique, process or way of doing something.

Methodology:

- It is a body of methods, rules, and postulates employed by a discipline, a particular procedure or set of procedures.
- Essentially a methodology is a collection of methods, practices, processes, techniques, procedures and rules.
- Examples: Waterfall Methodology, Agile Methodology, Scrum Methodology, Kanban Methodology, Scrum ban Methodology. etc.

Some Ways of Categorizing Software Projects:

- It is important to distinguish between the main types of software project because what is appropriate in one context might not be so in another.

1. Compulsory versus voluntary users: Point of Sales software has different requirement and user than Games.

2. Information System versus Embedded Systems: Information Systems helps in office day to day processes and embedded system are used to control machines.

- Examples of Information system are A stock control system, transaction processing system, knowledge management system, learning management system, database system etc.
- While Embedded systems control Microwave oven, AC, Mp3 Player, Digital Camera, Washing Machine etc.

3. Outsourced Project: Most of the companies outsource parts of its work to other companies. There can be several reasons to outsource project, e.g. company does not have sufficient expertise, or project is to be completed in limited period in that case it can be outsourced to company with their development centre.

- Outsourced project fetches one-time revenue in good amount while product based company project earns over period of time.

4. Object-Driven Development: Project also vary depending on the aim of the project. If aim of the project is to create product, then client has the responsibility for justifying the product. In that case Service level agreements are important as organisations contract out functions to external service suppliers.

Stakeholders:

- These are the people who have a stake or interest in the project.

Stakeholders can be categorised as:

1. Internal to project team: These will be under the direct managerial control of the project leader.

2. External to project team but within the same organization: For example, the project leader might need the assistance of the users to carry out systems and testing.

3. External to both project team and the organisation: External stakeholders may be customers (or users) who will benefit from the system that the project implements. They may be contractors who will carry out work for the project.

Setting Objectives:

- The objectives should define what the project team must achieve for project success.
- Although, different stakeholders have different motivations, the project objectives identify the shared intensions for the project.
- The objectives focus on the desired outcomes of the project rather than the tasks within it.
- Informally the objectives could be written as a set of statements following the opening words ‘the project will be a success if…’
- Thus one statement in a set of objectives might be ‘customers can order our products online’ rather than ‘to build an e-commerce website’.

- There may be several stakeholders, including users in different business areas, who might have some claim to project ownership.
- In such a case, a project authority needs to be explicitly identified with overall authority over the project.
- The authority is a project steering committee with overall responsibility for setting, monitoring and modifying the objects.
- The project manager runs the project on a day-to-day basis, but regularly reports to the steering committee.

Sub-objectives and goals:

- The mnemonic SMART is sometimes used to describe well defined objectives.

S-Specific:

- These are concrete and well defined.
- Ambiguous aspirations such as ‘to improve customer relations’ are unsatisfactory.
- Objectives should be defined so that it is obvious to all whether the project has been successful.

M-Measurable:

- There should be a measures of effectiveness which tell us how successful the project has been.
- For example: ‘to reduce customer complaints’ would be more satisfactory than ‘to improve customer relations’
- In some cases, be an answer to simple yes/no question. Ex: did we install new software by Dec 1?

A-Achievable:

- It must be within the power of the individual or group to achieve the objectives.

R-Relevant:

- The objective must be relevant to the true purpose of the project.

T-Time constrained:

- There should be a defined point in time by which the objective should have been achieved.

The Business Case:

- It is a document that allows decision-makers to determine whether the project is worth the investment.
- It presents a current business problem and suggests ways to solve it by implementing a certain initiative.
- It will quantify the project's cost and benefits.
- The benefits will be affected by the completion date. The sooner the project is completed; the sooner the benefits can be experienced.
- Example: A new web-based application might allow customers from all over the world to order a firm's products via the internet, increasing sales and thus increasing revenue and profits.

Any project plan must ensure that the business case is kept intact(un-touched).

- That development costs are not allowed to rise to a level which threatens to exceed the value of benefits.
- That the features of the system are not reduced to a level where the expected benefits cannot be realised.
- That the delivery date is not delayed so that there is an unaccepted loss of benefits.

Project Success and Failure:

Why Project is Success or Failure?

Project Success:

- Achievement of something desired, planned or attempted.
- It is also said that success is an event that accomplishes its intended purpose.

Criteria of a successful project:

- Clearly defined goals
- Sufficient resource allocation
- Adequate communication channels.
- Senior management support
- Client consultation.
- Technical tasks.
- Competent project manager and team members.

Factors that increase Success:

- **Top Management Support:** Lack of executive support can lead project failure so Positive Executive Support positively influence project outcome.
- **Client Involvement**
- **Experienced Project Manager**
- **Setting clear objectives**
- **Defined scope**

To summarize Success Criteria:

- You must define what success looks like for your project or else you won’t know if you have achieved it.
- Document the success criteria and get everyone to agree to them.
- Use continuous measurements where possible.
- Baseline today’s performance so you know where you are starting from
- Track as appropriate and report on your progress

Project Failure: A project that fails to perform a duty or an expected action, on-occurrence or non-performance.

Causes for Project Failures:

- **Lack of senior Management Support:** It happens when ‘Junior’ or Candidate PMs are allotted into Project and left unattended.
- **Poor Communication:** It causes delay or even failure since team members do not have the information they needed.
- **Inadequate resources:** Some tasks take longer than expected to complete, deadlines and milestones get missed.

- **No One is in control:** Not even the project manager, who is assigned to the project but not given the free hand to manage the project.
- **Poor definition of scope:** Project changes from its original objectives and goals. This can occur due to additional requirement from the client.
- **Poor Risk Management:** The project planning stages are not properly planned.
- **Incompetent project management skill:** Procedures and policies are not adhered to from inception.

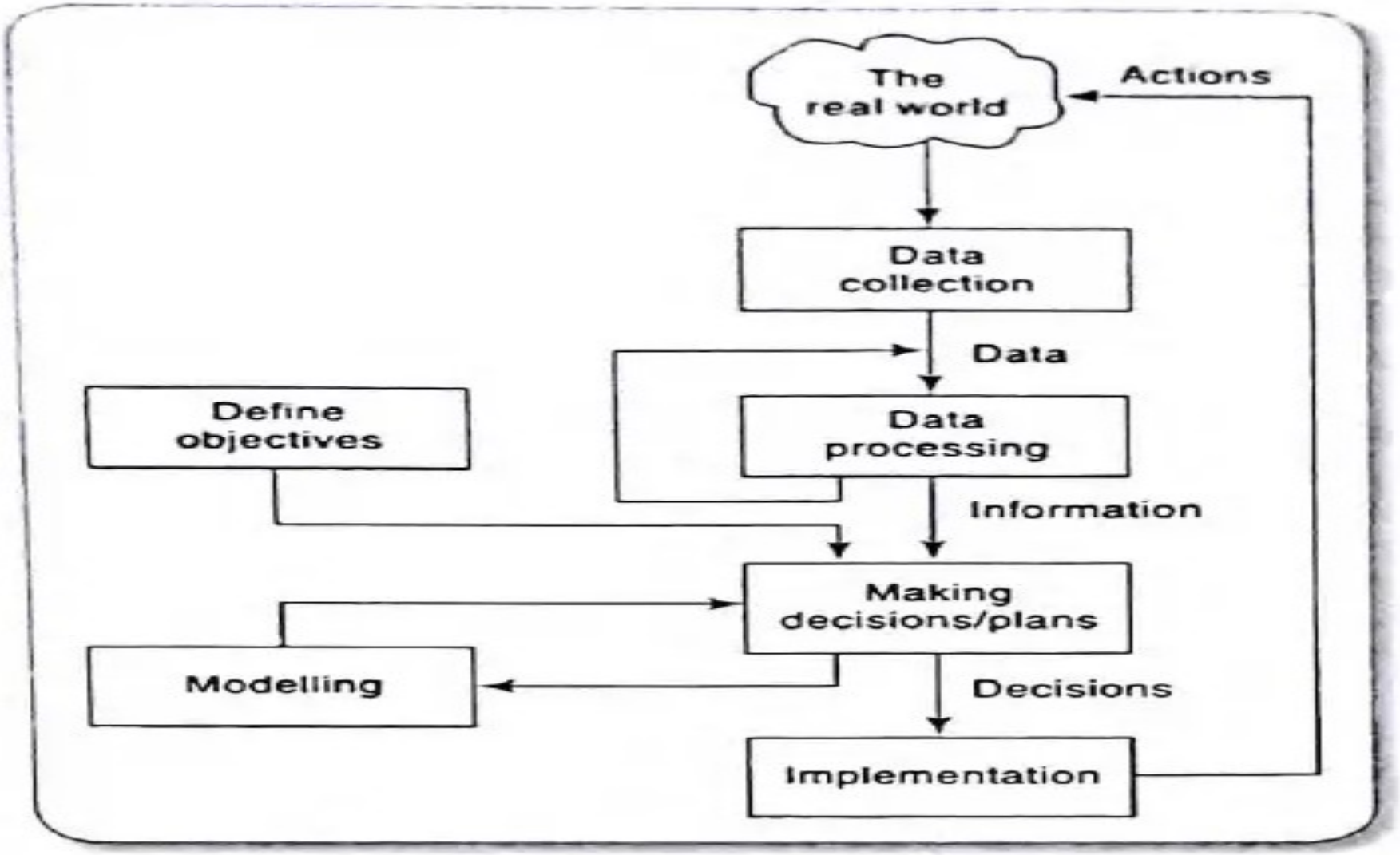
Management: defines all activities and tasks undertaken by one or more persons for the purpose of planning and controlling the activities of others in order to achieve objectives

Management Activities:

- **Planning**: deciding what is to be done
- **Organizing**: Making the arrangements
- **Staffing**: Selecting the right people for the job
- **Directing**: Giving instructions
- **Monitoring**: Checking on the progress
- **Controlling**: Taking actions to remedy hold-ups
- **Innovating**: Come up with solution when any problems emerge
- **Representing**: Liaising with clients, users, developers and other stakeholders.

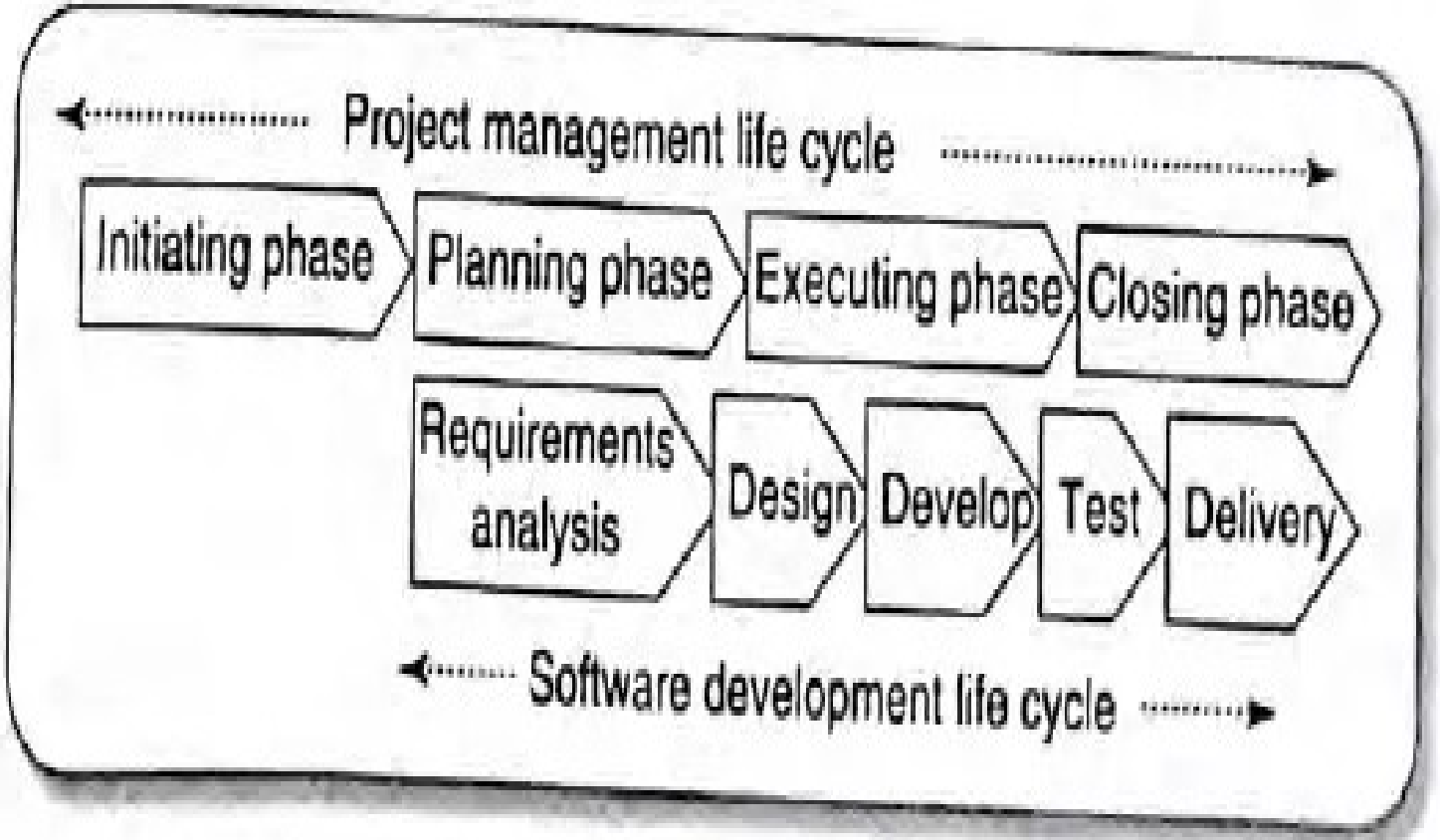
Management Control:

- It involves setting objectives for a system and then monitoring the performance of the system.



- Data Collection will be done by local manager.
- Data processing will be needed to transform raw data collected into useful information.
- To achieve the objectives, appropriate procedures have to be followed, this can be done by making decision plans.
- If there are any issues, solution has to be found out with the help of modelling.
- Having implemented the decision, the situation needs to be kept under review by collecting and processing further progress details.
- The project plan is dynamic and it will need constant adjustment during the project execution, hence action has to be performed continuously to make a good plan.

Project Management Process Life Cycle:



Different phases of project management life cycle and software development life cycle

1 Project Initiation:

- This is the starting period of your project when you should demonstrate the undertaking has value and is feasible.
- This stage incorporates making a business case, to legitimize the requirement for the undertaking, and an achievable study to show that it very well may be executed within a sensible time and cost.
- This is likewise an opportunity to make a task contract, a record that sets out precisely the thing the venture will convey.
- **Documentation**: Each project has documentation that should be finished before the undertaking can start vigorously.
- **Undertaking a feasibility study:** Identify the essential issue your task will tackle and whether your venture will convey an answer for that issue

- **Recognizing extension:** Define the profundity and broadness of the undertaking.
- **Assemble of the team:** You need resources to execute any project. Before you can make a project schedule, you need to create a project team with the skill sets and experience that the project demands.
- **Recognizing expectations:** Define the product or administration to provide.
- **Recognizing project partners:** Figure out whom the venture influences and what their requirements might be.
- **Building up a business case:** Use the above standards to think about the possible expenses and advantages of the task to decide whether it pushes ahead.
- **Building up an explanation of work:** Document the undertaking's goals, extensions, and expectations that you have distinguished already as a working understanding between the venture proprietor and those chipping away at the task.

2 Project Planning:

- In this phase, the project manager carries out several processes and creates the following documents.
- **Project Plan**: This identifies the project tasks, and a schedule for the project tasks that assigns project resources and time frames to the tasks.
- **Resource Plan**: It lists the resources, manpower and equipment that would be required to execute the project
- **Financial Plan**: It documents the plan for manpower, equipment and other costs.
- **Quality Plan:** Plan of quality targets and control plans are included in this document.
- **Risk Plan**: Lists the identification of the potential risks, their prioritization and a plan for the actions that would be taken to contain the different risks.

3 Project Execution:

- Which is the place where most of the work occurs. This is the stage where you complete the task exercises and achievements to create the expectations for the customer's or partner's fulfilment by following the arrangement made in the past stage.
- **Assignment Management:** To ensure an errand is done well, it must be dealt with each progression in transit, from wanting to consummation.
- This includes observing and answering to ensure the assignment is being executed within the period of the arranged timetable.
- Venture directors and colleagues need to deal with their undertakings. Assignment records and Kanban sheets are two mainstream apparatuses for the task of the board.

- **Timetable Management:** Whenever you've arranged a timetable, you need to screen it through the task execution to ensure it stays on target. A viable Schedule for the executives implies more prominent efficiency.
- You've to define objectives, needs, and cut-off times, presently as the undertaking errands are being executed, you must ensure those dates are coordinated with your timetable.
- **Cost Management:** Similarly, as you arranged your timetable, you arranged a financial plan. Yet, that doesn't mean your task is finished. As anybody with a wallet knows, cash tends to vanish. You need to control the venture expenses and keep them inside the concurred spending plan.

- **Quality Management:** You can create your expectations on schedule and inside the financial plan, yet if the quality is inadequate, the undertaking isn't effective. In this way, you need to ensure that you're meeting whatever quality prerequisites have been set by your partners.
- **Change Management:** Extensively, changing the board is an interaction for improving business measures, spending designation, and activities in an association. In any case, when applied to protect the board, the centre is limited to the actual undertaking and controlling changes in scope during the execution stage.
- **Acquirement Management:** Few is the undertaking that should be possible without buying, leasing, or agreement with outside assets. This interaction is called acquisition. Dealing with the different associations with merchants and providers is what is the issue here.

4 Project Closure:

- The fourth stage is project conclusion, in which the last expectations are introduced to the customer or partner.
- When affirmed, assets are delivered, documentation is finished and everything is approved. Now the venture supervisor and group can lead a posthumous to assess the exercises gained from the project and gain from the experience.
- Contingent upon the venture, the conclusion stage may likewise incorporate giving over control to an alternate group, for example, the tasks supervisory crew. In this situation, it is the work of the venture supervisor to guarantee that such a change happens easily.

- **Move Deliverables:** Your venture is tied in with creating a deliverable. That denotes the finish of the venture execution and the start of the task close. Subsequently, ensure you have all expectations recognized, total, and given off to the legitimate party.
- **Affirm Completion:** An undertaking isn't over until everybody sings. You need to get affirmation from all partners, customers, and even the group.
- That implies sign-offs so that there is no disarray and a minute ago change demands. Those close-down archives can be added to ProjectManager.com, either in the segment of the document or joined to the applicable assignment.
- Presently you have an advanced paper trail to settle on sure that everybody is in the arrangement.

- **Audit Documentation:** Typically, the venture supervisor is answerable for going over all agreements and documentation to ensure that all have been great and approved.
- Now and then in bigger associations, there is a devoted administrator for this work. Whoever does it, the significance of ensuring each one is spotted and t crossed couldn’t possibly be more significant.
- **Delivery Resources:** Before a task is truly done, you need to authoritatively deliver the group, any contractors, rentals, and so forth Have an interaction set up to advise and ensure everybody is settled up.

Traditional versus Modern Project Management Practices:

1.Planning Incremental Delivery:

- In olden days' projects were planned with sufficient details much before the actual project execution started.
- After the project initiation, monitoring and control activities work is carried out to ensure that project execution proceeded as per plan.
- The modern day projects are required to be completed in much shorter duration and rapid application development is used where the delivery of the project is done in different stages.

2 Quality Management:

- In olden days' customer used to adjust with the quality of the product but the awareness of the quality has now increased significantly.
- Hence the project manager has the responsibility to monitor the progress as well as the quality of the project.

3 Change Management:

- Earlier when the requirements were finalized by the customer any changes to the requirements were rarely entertained
- whereas now the customer suggestions are actively entertained and incorporated within the development process.

4. Release Management:

- It concerns planning, prioritizing, and controlling the different releases of a software.
- For almost every software, multiple release are made during its life cycle.
- Starting with an initial release, releases are made each time the code changes.
- Modern development processes such as the agile development processes advocate frequent and regular releases of the software to be made to the customer during the software development.

5. Risk Management:

- In modern software project management practices, effective risk management is considered very important to the success of a project.
- A risk is negative situation that may arise as the progresses and may threaten the success of the project.
- This involves identification of risks, assessment of the impacts of various risks, prioritization of risks and preparation of risk-containment plans.

Project Evaluation

It is a process of measuring the success of any project.

Cost, scope, Return On Investment (ROI), risks etc. It is carried at Three stages:

1.Pre-project evaluation: Covers project planning, gains confidence of all stakeholders, important for attracting investments.

2.On-going project evaluation: Project monitoring is done when project is running so that project proceeds as per set plans.

3.Post-project evaluation: It finds out what worked? And what went wrong ?

Importance of Project Evaluation

- Improves project performance
- Enables proper utilization of resources
- Revenue and profits also improve
- Enables proper budget allocation
- Reduces risks
- Areas of improvement are identified
- Improves team performance
- Improves confidence of all stakeholders

Evaluation of Individual Projects

Technical Assessment:

It consists of evaluating whether the required functionality can be achieved with current affordable technologies, Organizational policy, aimed at providing a consistent hardware/software infrastructure, is likely to limit the technical solutions considered.

The cost of the technology adopted must be taken into account in the cost-benefit analysis.

Cost-benefit analysis:

Even where the estimated benefits will exceed the estimated costs, it is often necessary to decide if the proposed project is the best of several options.

Cost-benefit analysis comprises two steps:

1. Identifying all of the costs and benefits of carrying out the project and operating the delivered application: These includes the development costs, the operating costs, and the benefits expected from the new system.

- ✓ Where the proposed system is a replacement, these estimates should reflect the change in costs and benefits due to the new system.
- ✓ A new sales order processing system, for example, could only claim to benefit an organization by the increase in sales due to the use of new system.

2.Expressing costs and benefits in common units: Must express each cost and benefits. The net benefit which is the difference between the two in money.

Most direct costs are easy to quantify in monetary terms and can be categorized as:

- **Development Costs:** Including development staff costs.
- **Setup Costs:** Consist of the costs of putting the system into place, mainly of any new hardware but also including the costs of file conversion, recruitment and staff training.
- **Operational Costs:** It relating to the system after installation.

Cash flow forecasting:

Estimating the overall costs and benefits of a project is producing a cash flow forecast which indicates when expenditure and income will take place.

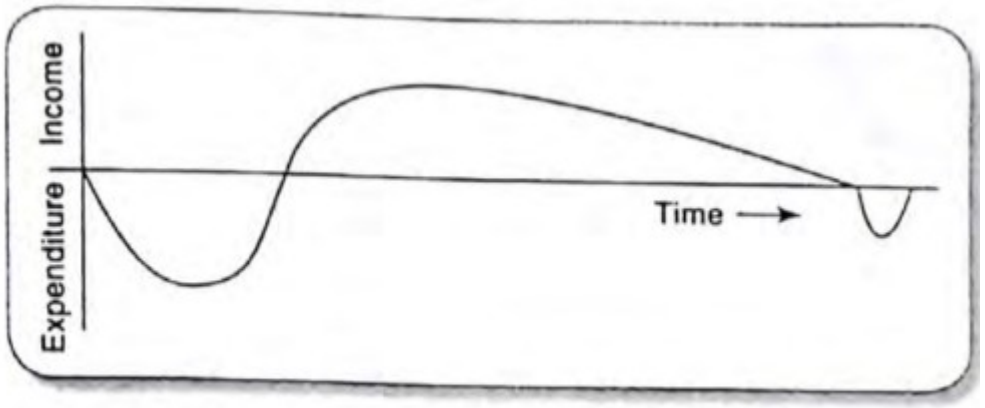


FIGURE 2.1 Typical product life cycle cash flow

- Need to spend money, such as staff wages, during a project's development.
- Such expenditure cannot wait until income is received.
- Need to know that fund this development expenditure either from the company's own resources or by borrowing.

- A forecast is needed of when expenditure, such as payment of salaries, and any income are to be expected.
- Accurate cash flow forecasting is difficult, as it is done early in the project's life cycle (at least before any expenditure is committed) and many items to be estimated (particularly the benefits of using software) might be some years in the future.

Cost-benefit Evaluation Techniques or Methods of Project Evaluation

1.Net Profit

It is the difference between the total costs and the total income over the life of the project.

2.Payback period

It is the time taken to break even or payback the initial investment.

Normally, the project with the shortest payback period will be chosen on the basis that an organization will wish to minimize the time that a project is ‘in debt’.

3.Return On Investment (ROI):

It is also known as Accounting Rate of Return (ARR), provides a way of comparing the net profitability to the investment required.

$$\text{ROI} = \frac{\textit{average annual profit}}{\textit{total investment}} \times 100$$

High ROI means good investment opportunity.

4. Net Present Value Method (NPV): It calculates difference between cash inflows and cash outflows.

$$\textit{Present value} = \frac{\textit{value in year } t}{(1+r)^t}$$

r-> Discount rate

Positive NPV indicates that project will generate more revenue.

Internal Rate of Return (IRR):

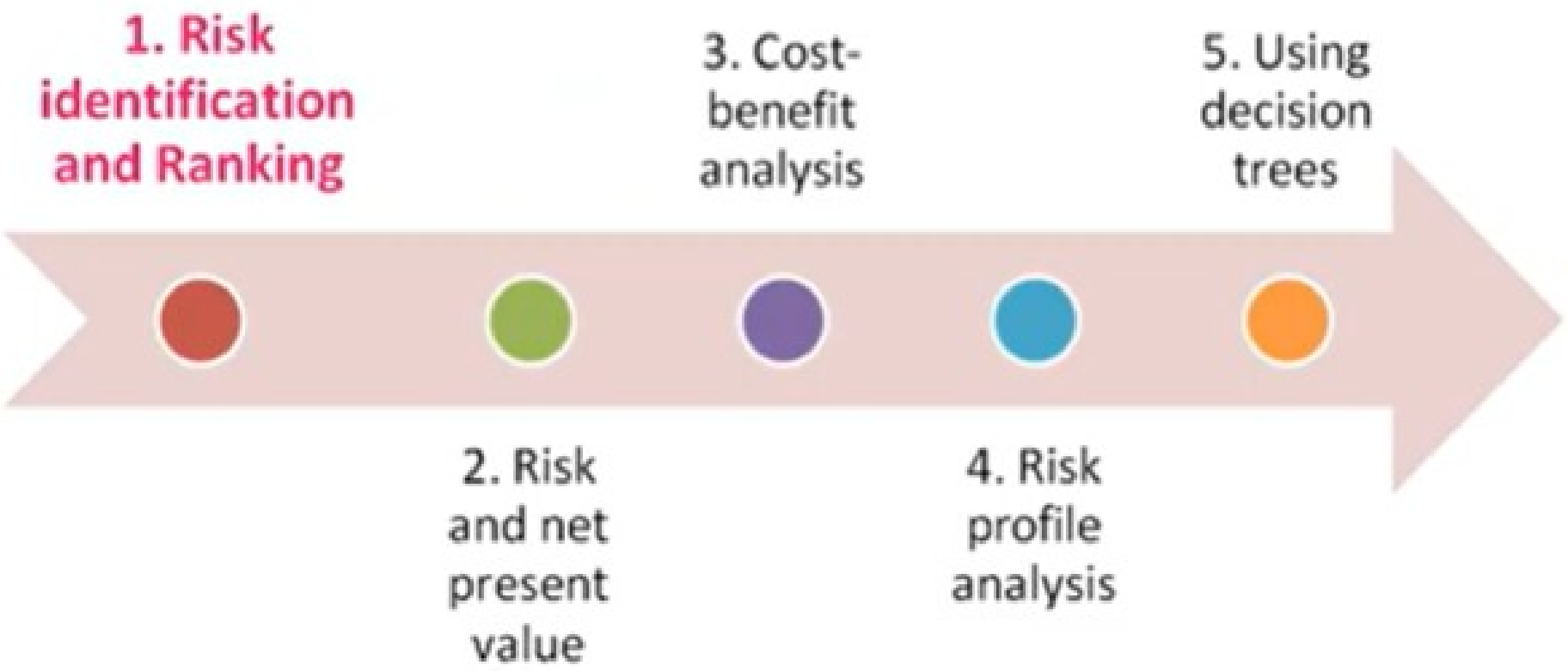
It is calculated as that percentage discount rate that would reduce an NPV of zero.

It is most easily calculated using a spreadsheet or other computer program that provides functions for calculating the IRR.

Risk Evaluation

Project Risk: Which prevents the project from being completed successfully.

Business Risk: Which prevents the project from being profitable.



1.Risk Identification and Ranking:

Approach: Constructing project risk matrix utilizing a checklist of possible risks and classify according to its relative importance and likelihood.

The table illustrate a basic project risk matrix listing some of the business risks for a project, with their importance and likelihood classified as high (H), medium(M),low(L) or exceedingly unlikely(_)

Project risk matrix for an e-commerce application:

Risk	Importance	Likelihood
Client rejects proposed look and feel of site	H	—
Competitors undercut prices	H	M
Warehouse unable to deal with increased demand	M	L
Online payment has security problems	M	M
Maintenance costs higher than estimated	L	L
Response times deter purchasers	M	M

The project risk matrix may be used as a way of evaluating projects (those with high risks being less favored) or as a means of identifying and ranking the risks for a specific project.

2. Risk and Net Present Value:

- If project is relatively risks, then use higher discount rate to calculate NPV
- Reasonably safe project: Additional 2% risk premium
- Fairly risky project: additional 5% risk premium
- Risks are generally categorized using scoring method and risks premiums can be designated for each category some means of considering risk.

3. Cost Benefit Analysis:

- It is required to consider each possible outcome
- Estimate the probability of its occurring and the corresponding value of the outcome
- Instead of single cash flow, consider set of cash flow forecasts
- Each with an associated probability of occurring
- The value of the project is then obtained by summing the cost or benefit for each possible outcome weighted by its corresponding probability

A payroll project's cost-benefit analysis by company X

Sales	Annual sales income (£)	Probability	Expected value (£)
	i	p	$i \times p$
High	800,000	0.1	80,000
Medium	650,000	0.6	390,000
Low	100,000	0.3	30,000
Expected Income			500,000

4. Risk Profile Analysis:

Approach: Attempts to overcome some of the objections to cost-benefit averaging is the construction of risk profiles using sensitivity analysis.

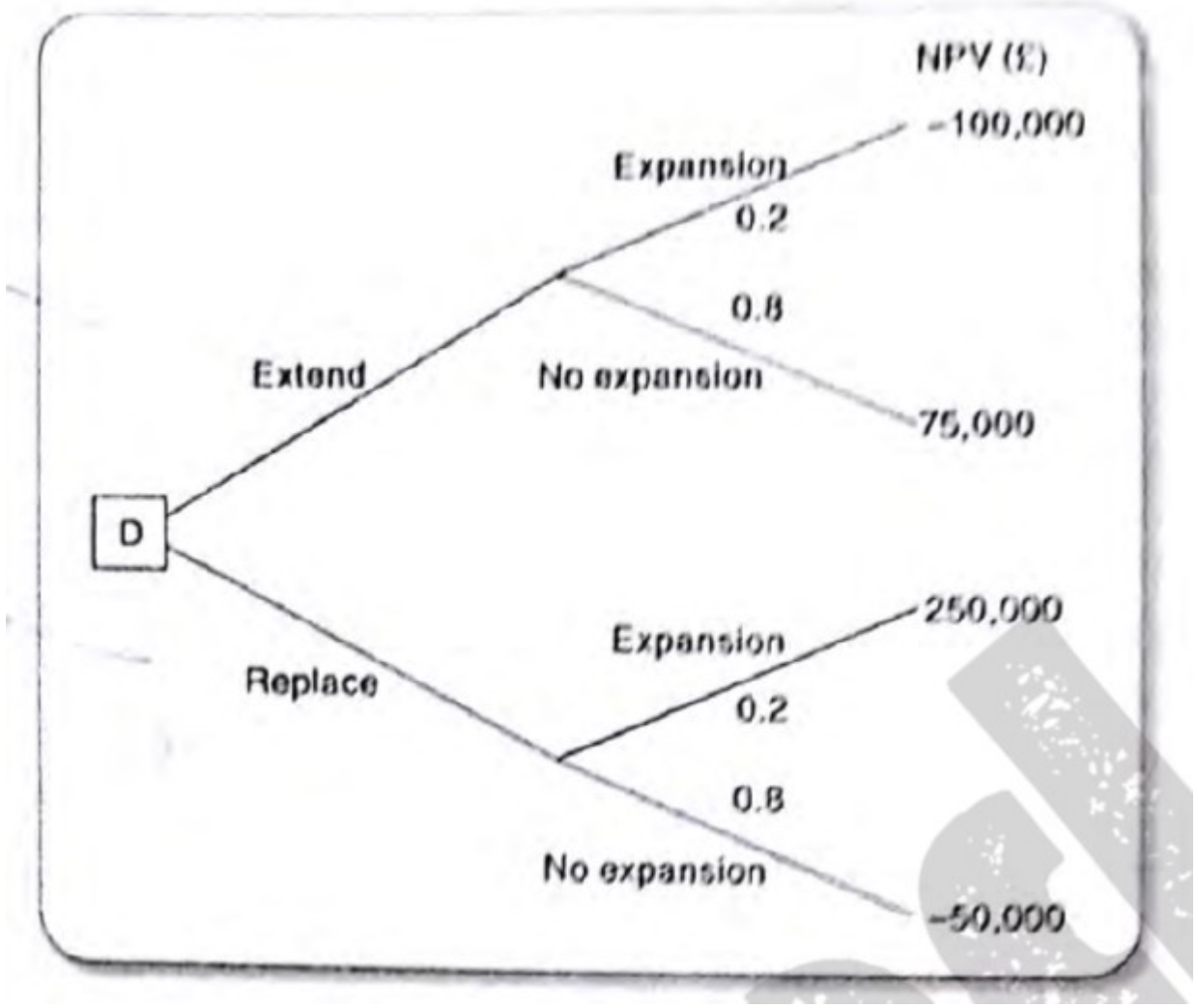
This involves varying each of the parameters that affect the project's cost or benefit to ascertain how sensitive the project's profitability is to each factor.

For example, vary one of our original estimates by plus or minus 5% and recalculate the expected costs and benefits for the project.

By repeating this exercise for each of our estimates in turn we can evaluate the sensitivity of the project to each factor.

By studying the results of a sensitivity analysis, we can identify those factors that are most important to the success of the project.

We then need to decide whether we can exercise greater control over them or otherwise mitigate their effects.

If neither is the case, then we must live with the risk or abandon the project.

5. Using Decision Trees: